

Tech & Teach

Digital Chip Design Program

RTL Design

Task #11

Farah Alyahya

Rana Alsubaihi

Joud Almshari

Tala Alrsheed

Hoor Alkhedairi

Horiah Alqufaydi

Sandra Alomar

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Seq & Comb Circuits Datapath – Part 2

Version 1.0

Task #1.1

1.1.5

1- Clock Divider (clockDivider):

Purpose: Generates three lower-frequency clock signals ($F/2$, $F/4$, and $F/8$) from the main input clock signal (F).

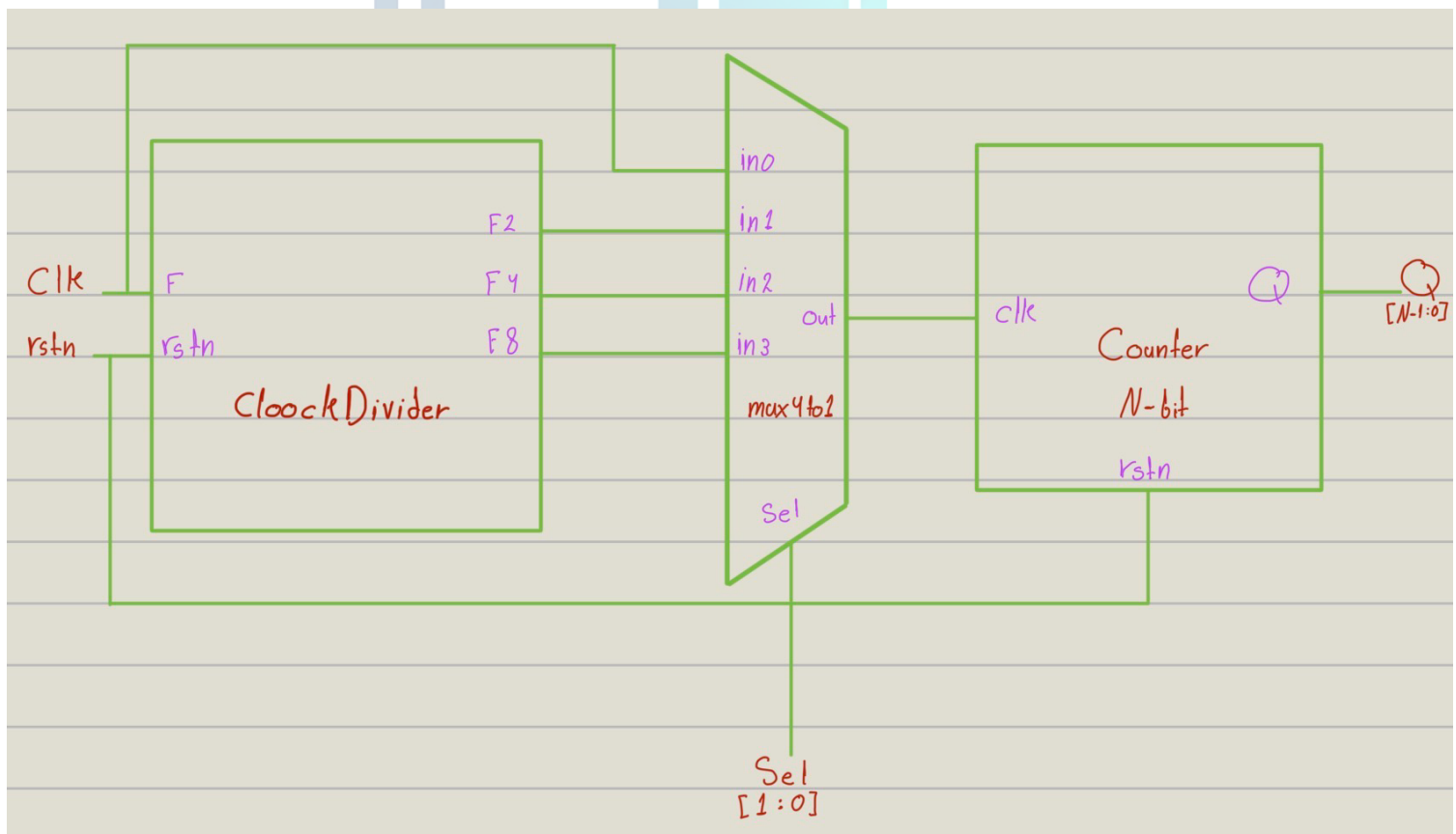
2 - 4-to-1 Multiplexer (mux4to1):

Purpose: Selects one of the four clock signals (F , $F/2$, $F/4$, OR $F/8$) based on the 2-bit speed selection input $sel[1:0]$.

3- N-bit Counter (counter):

Purpose: Increments the counter output $Q[N-1 : 0]$ on every rising edge of the selected clock frequency.

1.1.6



1.1.7

```
1 module counter
2   #(parameter N = 4) (
3     input clk,rstn,
4     output reg [N-1:0] Q
5   );
6   always_ff @(posedge clk or negedge rstn) begin
7     if (!rstn)
8       Q <= '0;
9     else
10      Q <= Q + 1;
11   end
12 endmodule
```



```
1 module mux4to1 (
2   input in0,in1,in2,in3,
3   input [1:0] sel,
4   output logic out
5 );
6
7   always_comb begin
8     case (sel)
9       2'b00: out = in0;
10      2'b01: out = in1;
11      2'b10: out = in2;
12      2'b11: out = in3;
13      default: out = in0;
14    endcase
15  end
16 endmodule
```

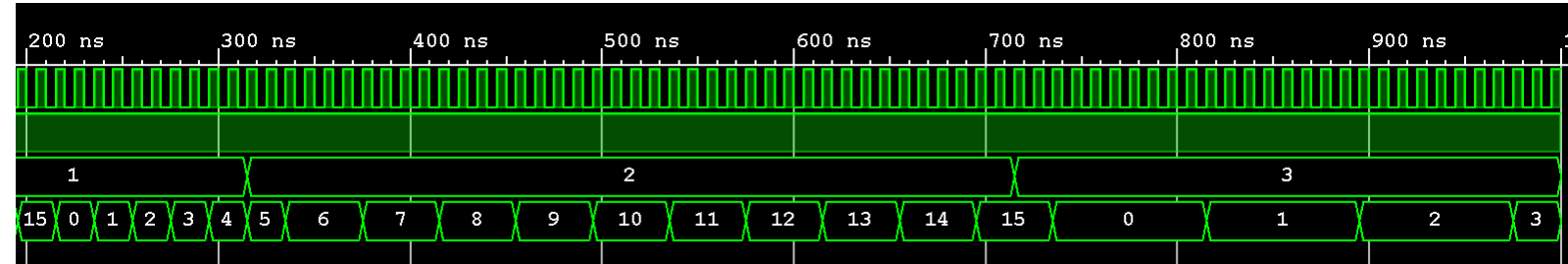
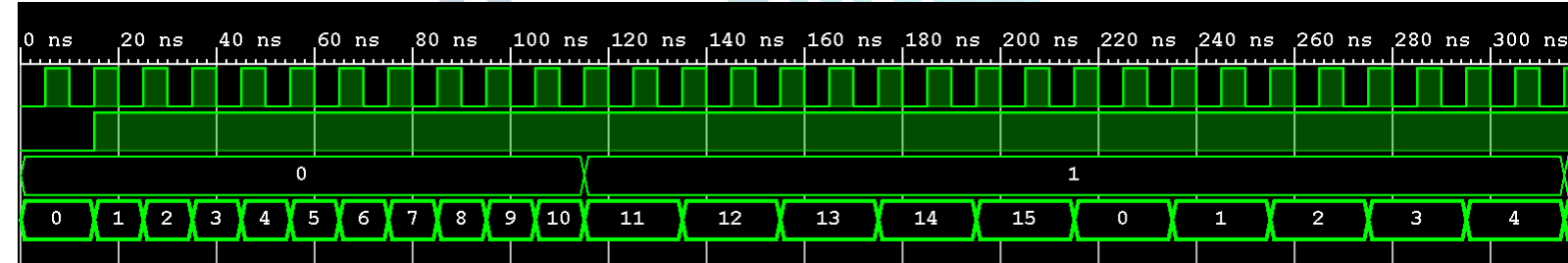
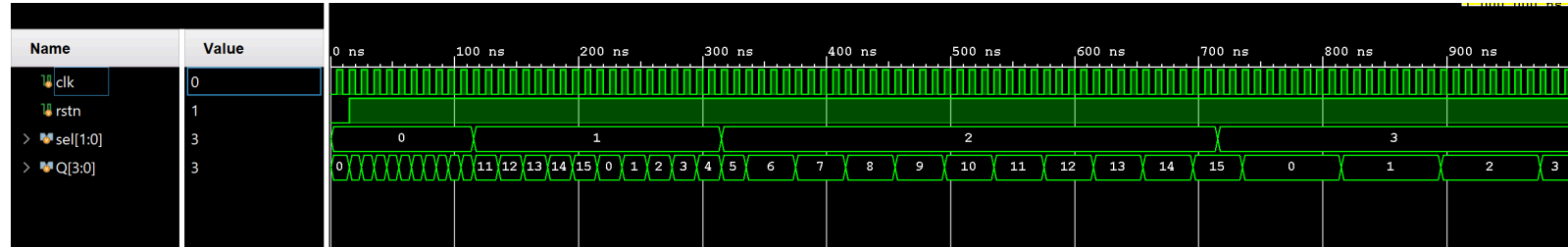
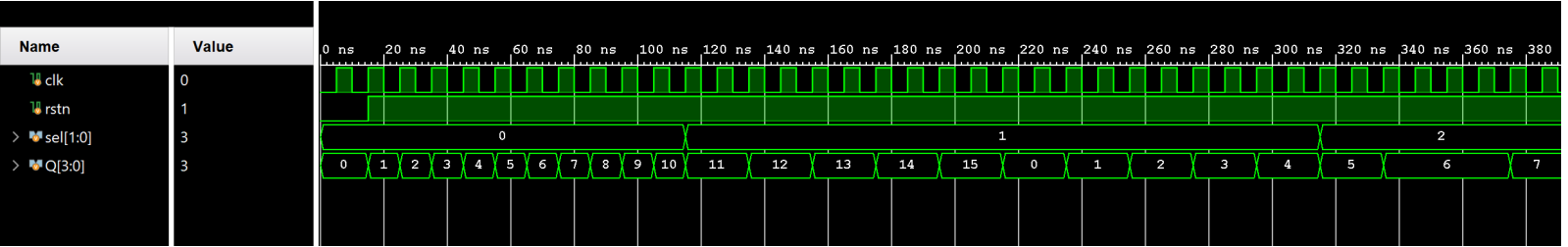
```

1  module CounterSpeedController
2      #(parameter N = 4) (
3          input  clk,rstn,
4          input  [1:0] sel,
5          output [N-1:0] Q
6      );
7
8      logic F2, F4, F8;
9      logic selected_clk;
10
11     clockDivider clkDiv (
12         .F(clk),
13         .rstn(rstn),
14         .F2(F2),
15         .F4(F4),
16         .F8(F8)
17     );
18
19     mux4to1 clkMux (
20         .in0(clk),
21         .in1(F2),
22         .in2(F4),
23         .in3(F8),
24         .sel(sel),
25         .out(selected_clk)
26     );
27
28     counter #(N) Counter (
29         .clk(selected_clk),
30         .rstn(rstn),
31         .Q(Q)
32     );
33
34  endmodule

```

1.1.8

```
1  `timescale 1ns / 1ps
2  module CounterSpeedController_tb;
3      parameter N = 4;
4
5      logic clk,rstn;
6      logic [1:0] sel;
7      logic [N-1:0] Q;
8
9      CounterSpeedController #(N) dut (
10         .clk(clk),
11         .rstn(rstn),
12         .sel(sel),
13         .Q(Q)
14     );
15
16     always #5 clk = ~clk;
17
18     initial begin
19         clk  = 0;
20         rstn = 0;
21         sel  = 2'b00;
22
23         #15;
24         rstn = 1;
25
26         #100;
27         sel = 2'b01;
28
29         #200;
30         sel = 2'b10;
31
32         #400;
33         sel = 2'b11;
34
35         #800 $finish;
36     end
37 endmodule
```



Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1	✓	✓	✓